



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICAL LITERACY P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Mathematical Literacy Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark (M = method, A = accuracy). Give follow-through marks (CA) where a learner correctly uses an earlier wrong answer. Penalise once for missing units or the R sign.

QUESTION 1

- 1.1 $2\,500 + 850 + 1\,400 + 299 + 500 + 450 + 380 \checkmark = R6\,379 \checkmark$ (2)
- 1.2 $6\,800 - 6\,379 \checkmark = R421 \checkmark$ (2)
- 1.3 $\frac{2\,500}{6\,800} \times 100 \checkmark\checkmark = 36,76\% \checkmark$ (3)
- 1.4 $2\,500 \times 8\% = R200 \checkmark$; $2\,500 + 200 \checkmark = R2\,700 \checkmark$ (or $2\,500 \times 1,08$) (3)
- 1.5 $421 - 200 = R221 \checkmark$; yes, she still has R221 left ✓ (2)
- 1.6 $158 \times R2,85 \checkmark = R450,30 \checkmark$ (2)
- 1.7 Price without VAT = $345 \div 1,15 \checkmark = R300 \checkmark$; VAT = $345 - 300 = R45 \checkmark$ (3)
- 1.8.1 $149 \times 24 \checkmark = R3\,576 \checkmark$ (2)
- 1.8.2 $80 \times 24 = R1\,920 \checkmark$; $1\,999 + 1\,920 \checkmark = R3\,919 \checkmark$ (3)
- 1.8.3 Option A ✓, by $3\,919 - 3\,576 = R343 \checkmark$ (2)
- 1.9 Any ONE sensible reason, e.g. for emergencies such as illness or job loss ✓✓ / to buy something big later without debt / to earn interest. (2)

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QUESTION 2

- 2.1 2.1.1: $2\,500 \text{ m} \checkmark$ 2.1.2: $0,75 \ell \checkmark$ 2.1.3: $(3 \times 60) + 45 = 225 \text{ min} \checkmark$ 2.1.4: $1,45 \text{ kg} \checkmark$ (4)
- 2.2.1 $8 \times 6,5 \checkmark = 52 \text{ m}^2 \checkmark$ (2)
- 2.2.2 Area of one tile = $0,5 \times 0,5 = 0,25 \text{ m}^2 \checkmark$; $52 \div 0,25 \checkmark = 208 \text{ tiles} \checkmark$ (3)
- 2.2.3 $208 \div 10 = 20,8 \checkmark \rightarrow 21 \text{ boxes (round up)} \checkmark$; $21 \times R189 \checkmark = R3\,969 \checkmark$ (4)
- 2.2.4 $2 \times (8 + 6,5) \checkmark = 29 \text{ m} \checkmark$ (2)
- 2.2.5 $29 \times R35 \checkmark = R1\,015 \checkmark$ (2)
- 2.3.1 $3,142 \times 0,7^2 \times 1,5 \checkmark\checkmark = 2,31 \text{ m}^3 \checkmark$ (3)
- 2.3.2 $2,30937 \times 1\,000 \checkmark = 2\,309,37 \ell$ (accept $2\,310 \ell$) ✓ (2)
- 2.4 $06:45 + 2 \text{ h} = 08:45 \checkmark$; $+ 50 \text{ min} = 09:35 \checkmark$ (2)
- 2.5 $(1,8 \times 25) + 32 \checkmark = 77 \text{ }^\circ\text{F} \checkmark$ (2)

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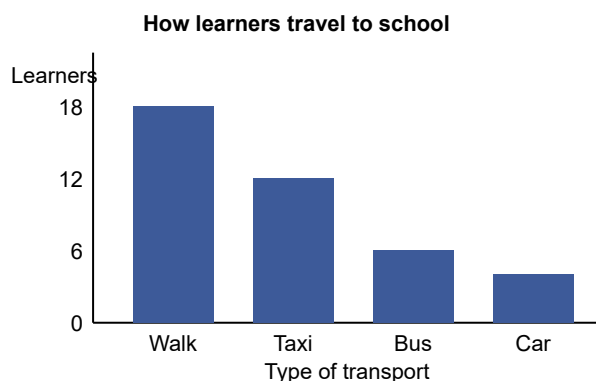
QUESTION 3

- 3.1 1 unit (cm) on the map ✓ represents 50 000 of the same units (cm) in reality ✓. (2)
- 3.2 $4,2 \times 50\,000 = 210\,000\text{ cm}$ ✓; $\div 100\,000$ ✓ = 2,1 km ✓ (3)
- 3.3 $2,1 \div 5 = 0,42\text{ h}$ ✓; $0,42 \times 60$ ✓ = 25,2 min (about 25 minutes) ✓ (3)
- 3.4 D4 ✓ (1)
- 3.5 The shop ✓ (1)
- 3.6 South-east ✓✓ (2)
- 3.7 $3,5\text{ cm} \times 100 = 350\text{ cm} = 3,5\text{ m}$ ✓; $4\text{ cm} \times 100 = 4\text{ m}$ ✓; area = $3,5 \times 4$ ✓ = 14 m^2 ✓ (4)
- 3.8 Yes ✓. 1 : 50 000 is a larger scale: it shows a smaller area, so more detail can be shown ✓. (2)
- 3.9 So that real distances can be worked out from the map ✓✓ (accept: to show how much the area was made smaller). (2)

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QUESTION 4

- 4.1 19 ; 23 ; 27 ; 28 ; 30 ; 35 ; 35 ; 35 ; 41 ; 44 ; 47 ✓✓ (2)
- 4.2 35 ✓ (1)
- 4.3 The 6th value (middle of 11) ✓ = 35 ✓ (2)
- 4.4 Total = 364 ✓; $364 \div 11$ ✓ = 33,09 ✓ (3)
- 4.5 $47 - 19$ ✓ = 28 ✓ (2)
- 4.6 50% of 50 = 25 marks ✓; learners below 25: 19 and 23, so 2 failed ✓ (5)
- 9 passed ✓; $\frac{9}{11} \times 100$ ✓ = 81,82% ✓
- 4.7 Median ✓. It is not pulled down by the low mark of 19 (outlier); it is the middle mark ✓. (2)
- (Accept "mean" with a sensible reason, e.g. it uses all the marks.)
- 4.8.1 (5)



Marks: heading ✓; both axes labelled ✓; even scale ✓; correct bar heights (all four) ✓✓

4.8.2 $\frac{18}{40} \times 100 \checkmark = 45\% \checkmark$ (2)

4.8.3 $\frac{12}{40} \checkmark = 0,3 \checkmark$ (2)

4.9 Discrete data can only take certain values (counted), e.g. number of learners ✓. (2)
Continuous data can take any value in a range (measured), e.g. height or mass ✓.

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TOTAL: 100

END OF MARKING GUIDELINE